RBMK-1000 Simulation

Physics Equations Reference

This document lists every equation used in the simulation, grouped by subsystem. Equations with standard physical derivations are noted as such. Constants that are hand-tuned to produce qualitatively correct behaviour (rather than derived from nuclear data) are flagged in the constants table (Table 2).

1 Neutron Motion

Neutrons move as free particles between interactions. Each timestep Δt the position is updated by a simple Euler step:

$$x(t + \Delta t) = x(t) + v_x \, \Delta t, \qquad y(t + \Delta t) = y(t) + v_y \, \Delta t \tag{1}$$

To keep the spatial error below one collision radius, the timestep is automatically subdivided into n_{sub} substeps:

$$n_{\text{sub}} = \left\lceil \frac{v_{\text{fast}} \, \Delta t}{3} \right\rceil, \qquad \Delta t_{\text{sub}} = \frac{\Delta t}{n_{\text{sub}}} \tag{2}$$

2 Interaction Probabilities

All interactions are modelled as independent Poisson processes. For a process with rate λ (events per second), the probability of at least one event in a short interval Δt is:

$$P(\text{interact in } \Delta t) = \lambda \, \Delta t \qquad (\text{valid for } \lambda \, \Delta t \ll 1) \tag{3}$$

Equation (3) is applied by drawing a uniform random number $u \sim \mathcal{U}(0, 1)$ and triggering the event when $u < \lambda \, \Delta t$.

Interaction	**Condition for removal / change**	**Rate / probability**
Graphite moderation (fast $\to$ thermal)	$u < \lambda_G \, \Delta t$	$\lambda_G = 9.0 \text{ s}^{-1}$
Water moderation (fast $\to$ thermal)	$u < \lambda_{W,\text{mod}} \, \Delta t$	$\lambda_{W,\text{mod}} = 1.2 \text{ s}^{-1}$
Water absorption	$u < \lambda_{W,\text{abs}} \, \Delta t$	$\lambda_{W,\text{abs}} = 1.7 \text{ s}^{-1}$

Interaction	**Condition for removal / change**	**Rate / probability**
Coolant absorption (between fuel pins, void-reduced)	$u < \lambda_c\,\Delta t$	See Eq. (4)
Thermal fission (on reactive fuel dot)	$u < P_{\text{th}}$	$P_{\text{th}} = 0.92$
Fast fission (on reactive fuel dot)	$u < P_{\text{fast}}$	$P_{\text{fast}} = 0.05$
Xenon absorption (thermal only)	thermal neutron enters xenon dot	certain (probability 1)
Xenon decay	$u < \lambda_{\text{Xe}}\,\Delta t$	$\lambda_{\text{Xe}} = 0.04\ \text{s}^{-1}$
Fuel regeneration	$u < \lambda_{\text{regen}}\,\Delta t$ when fissile fraction below target	$\lambda_{\text{regen}} = 0.15\ \text{s}^{-1}$

3 Void-Dependent Coolant Absorption

The key mechanism of the positive void coefficient: as the void fraction f_v rises (water boils to steam), the effective coolant absorption rate falls linearly, increasing reactivity. Inserted boron rods on either side of the channel add their own absorption term weighted by $n_{\text{rod}} \in \{0, 1, 2\}$, the number of bordering rods whose boron section overlaps the neutron height:

$$\lambda_c = \lambda_{\text{fc}}\,(1 - f_v) + \lambda_{\text{rod}}\,n_{\text{rod}} \tag{4}$$

where $\lambda_{\text{fc}} = 1.9\ \text{s}^{-1}$ is the full-water coolant absorption rate and $\lambda_{\text{rod}} = 8.5\ \text{s}^{-1}$ is the per-rod boron worth over adjacent fuel.

4 Coolant Dynamics

4.1 Fission flux per channel

Each fission in channel i increments a per-channel flux counter φ_i. Between frames the flux decays exponentially (representing the finite thermal response time of the coolant):

$$\varphi_i(t + \Delta t) = \varphi_i(t)\ \exp\!\left(-\frac{\Delta t}{\tau_\varphi}\right) \tag{5}$$

with $\tau_\varphi = 0.5$ s.

4.2 Channel temperature

The coolant temperature T_i in channel i follows a first-order forced cooling ODE (Newton's law of cooling with a fission heat source). Integrated with forward Euler:

$$T_i(t + \Delta t) = T_i(t) + H\,\varphi_i(t)\,\Delta t - C\,T_i(t)\,\Delta t, \qquad T_i \geq 0 \tag{6}$$

where $H = 0.035$ is the heat deposited per fission event (dimensionless simulation units) and $C = 0.5\ \text{s}^{-1}$ is the cooling rate.

4.3 Void fraction

The void fraction is a linear ramp from zero at the boiling threshold to unity at the full-void temperature, clamped to $[0, 1]$:

$$f_{v,i} = \operatorname{clamp}\left(\frac{T_i - T_{\text{boil}}}{T_{\text{full}} - T_{\text{boil}}},\ 0,\ 1\right) \tag{7}$$

with $T_{\text{boil}} = 0.45$ and $T_{\text{full}} = 1.3$ (dimensionless simulation temperature units).

The average void fraction reported in the diagnostics panel is:

$$\bar{f}_v = \frac{1}{N_{\text{ch}}} \sum_{i=1}^{N_{\text{ch}}} f_{v,i} \tag{8}$$

5 Reactor Diagnostics

5.1 Exponential moving average

Raw per-frame counts are too noisy to display directly. Each diagnostic quantity x is smoothed over a window of $\tau_{\text{EMA}} = 0.6$ s:

$$\widetilde{x}(t + \Delta t) = \widetilde{x}(t)\,\alpha + x(t), \qquad \alpha = \exp\left(-\frac{\Delta t}{\tau_{\text{EMA}}}\right) \tag{9}$$

This is applied to fission-neutron counts, total fission counts, and neutron loss counts independently.

5.2 Effective multiplication factor

k_{eff} is estimated from the smoothed neutron balance (Monte Carlo analogue of the four-factor formula):

$$k_{\text{eff}} = \frac{\widetilde{N}_{\text{fiss-n}}}{\widetilde{N}_{\text{fiss}} + \widetilde{N}_{\text{loss}}} \tag{10}$$

where $\widetilde{N}_{\text{fiss-n}}$ is the smoothed count of fission-born neutrons (i.e. ν per fission), $\widetilde{N}_{\text{fiss}}$ is the smoothed fission count, and $\widetilde{N}_{\text{loss}}$ is the smoothed count of neutrons lost to absorption or leakage without causing fission.

5.3 Reactivity and dollars

The reactivity ρ and its expression in dollars \$ are standard reactor-physics definitions [1]:

$$\rho = \frac{k_{\text{eff}} - 1}{k_{\text{eff}}} \tag{11}$$

$$\rho\,[\$] = \frac{\rho}{\beta_{\text{eff}}} \tag{12}$$

The value $\beta_{\text{eff}} = 0.0065$ is taken directly from INSAG-7 [1] and is the only parameter in the simulation taken from a published nuclear-data source.

5.4 Criticality classification

$$\text{State} = \begin{cases} \text{Subcritical} & \rho\,[\$] < -0.02 \\ \text{Critical} & |\rho\,[\$]| \le 0.02 \\ \text{Supercritical} & \rho\,[\$] > 0.02 \end{cases}$$

6 Constants

Table 2: Simulation constants. *Source*: "real" = taken from published nuclear data; "tuned" = hand-adjusted to produce qualitatively correct behaviour.

Symbol	**Value**	**Unit**	**Description**	**Source**
β_{eff}	0.0065	—	Effective delayed-neutron fraction	Real (INSAG-7 [1])
ν	2–3	—	Neutrons emitted per fission (mean ≈ 2.5)	Real (^{235}U: 2.43 [3])
P_{th}	0.92	—	Thermal fission probability per collision	Tuned
P_{fast}	0.05	—	Fast fission probability per collision	Tuned
λ_G	9.0	s^{-1}	Graphite moderation rate (fast$\rightarrow$thermal)	Tuned
$\lambda_{W,\text{mod}}$	1.2	s^{-1}	Water moderation rate	Tuned
$\lambda_{W,\text{abs}}$	1.7	s^{-1}	Water absorption rate	Tuned
λ_{fc}	1.9	s^{-1}	Coolant absorption rate (full water, no void)	Tuned
λ_{rod}	8.5	s^{-1}	Per-rod boron absorption worth over adjacent fuel	Tuned
λ_{Xe}	0.04	s^{-1}	Xenon-135 decay rate	Tuned
λ_{regen}	0.15	s^{-1}	Fuel regeneration / xenon-breed rate	Tuned
H	0.035	—	Heat deposited per fission (sim. units)	Tuned
C	0.5	s^{-1}	Coolant cooling rate	Tuned
τ_φ	0.5	s	Fission-flux decay time constant	Tuned
T_{boil}	0.45	—	Boiling threshold (sim. temperature units)	Tuned
T_{full}	1.3	—	Full-void temperature (sim. temperature units)	Tuned
τ_{EMA}	0.6	s	Diagnostic smoothing window	Tuned
v_{fast}	260	px s^{-1}	Fast neutron speed	Tuned
v_{thermal}	120	px s^{-1}	Thermal neutron speed	Tuned
$\dot{S}$	9.0	s^{-1}	Spontaneous fission source rate	Tuned
f^*_{react}	0.28	—	Target equilibrium fissile fuel fraction	Tuned

Symbol	Value	Unit	Description	Source
Y_{Xe}	0.05	—	Xenon poison yield per fission	Tuned
Y_{burn}	0.12	—	Burnup yield per fission	Tuned

References

References

[1] International Atomic Energy Agency, *The Chernobyl Accident: Updating of INSAG-1*, INSAG-7, IAEA Safety Series No. 75-INSAG-7, Vienna, 1992.
https://www-pub.iaea.org/MTCD/publications/PDF/Pub913e_web.pdf

[2] Fröhner, A. *et al.*, "A simplified analysis of the Chernobyl accident," *EPJ Nuclear Sciences & Technologies*, 7, 2021.
https://doi.org/10.1051/epjn/2020018

[3] World Nuclear Association, *RBMK Reactors*, Appendix.
https://world-nuclear.org/information-library/appendices/rbmk-reactors

Summary of All Equations

#	Name	Equation	Description
1	Neutron position update	$\mathbf{r}(t{+}\Delta t) = \mathbf{r}(t) + \mathbf{v}\,\Delta t$	Euler integration; applied per substep
2	Integration substeps	$n_{\text{sub}} = \left\lceil \dfrac{v_{\text{fast}}\,\Delta t}{3} \right\rceil$	Keeps spatial error below one collision radius
3	Poisson interaction	$P(\text{event in }\Delta t) = \lambda\,\Delta t$	Basis for all stochastic interaction tests
4	Void-dependent absorption	$\lambda_c = \lambda_{\text{fc}}(1 - f_v) + \lambda_{\text{rod}}\,n_{\text{rod}}$	**The positive void coefficient**: lower $f_v \Rightarrow$ less absorption $\Rightarrow$ more power
5	Fission flux decay	$\varphi_i(t{+}\Delta t) = \varphi_i(t)\,e^{-\Delta t/\tau_\varphi}$	Models thermal lag of coolant heating
6	Coolant temperature	$T_i \mathrel{+}= H\varphi_i\,\Delta t - C\,T_i\,\Delta t$	Newton cooling + fission heat source
7	Void fraction	$f_{v,i} = \operatorname{clamp}\!\left(\dfrac{T_i - T_{\text{boil}}}{T_{\text{full}} - T_{\text{boil}}},\ 0,\ 1\right)$	Linear ramp from boiling to full steam
8	Average void fraction	$\bar{f}_v = \dfrac{1}{N_{\text{ch}}}\displaystyle\sum_{i=1}^{N_{\text{ch}}} f_{v,i}$	Reported as *Steam void %* in diagnostics
9	Exponential moving average	$\tilde{x} \leftarrow \tilde{x}\,e^{-\Delta t/\tau_{\text{EMA}}} + x$	Smooths per-frame neutron counters
10	Effective multiplication factor	$k_{\text{eff}} = \dfrac{\widetilde{N}_{\text{fiss-n}}}{\widetilde{N}_{\text{fiss}} + \widetilde{N}_{\text{loss}}}$	Monte Carlo neutron balance; critical at $k_{\text{eff}} = 1$
11	Reactivity	$\rho = \dfrac{k_{\text{eff}} - 1}{k_{\text{eff}}}$	Standard reactor-physics definition
12	Reactivity in dollars	$\rho\,[\$] = \dfrac{\rho}{\beta_{\text{eff}}},\quad \beta_{\text{eff}} = 0.0065$	β_{eff} from INSAG-7; only real-data constant